

Generative Digital Twin Methods for Spacecraft Telemetry Anomaly Detection: Toward Data-Driven FDIR Design and Testing

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Python · PyTorch · GANs / VAEs / LSTMs · Anomaly Detection · Digital Twins · MATLAB/Simulink

The problem

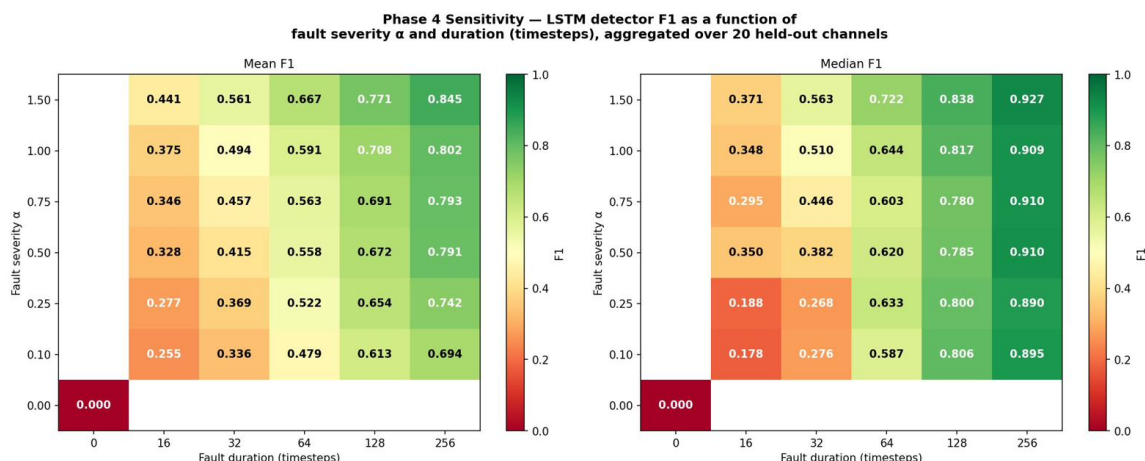
Onboard Fault Detection, Isolation and Recovery (FDIR) is among the most safety-critical software a spacecraft carries, and the flight-standard approach is model-based: a fault is flagged when a residual crosses a manually tuned threshold. This has a structural blind spot — **contextual faults** that distort a signal's temporal pattern while its amplitude stays within limits raise no alarm, yet they account for 43 of the 105 anomalies in NASA's SMAP/MSL in-orbit benchmark. This thesis asks whether data-driven methods can close that gap, and whether generative models can supply the scarce fault data needed to build, calibrate and certify the detectors themselves.

Approach and key results

- **Data-driven detection on real telemetry.** Variational-Autoencoder and LSTM detectors, trained on real NASA SMAP/MSL telemetry, raised mean contextual-fault F1 from the classical baseline's 0.052 to 0.836 (LSTM) — a sixteenfold gain — and independently reproduced the published LSTM benchmark to within 0.006.
- **Generative fault synthesis.** A conditional Wasserstein GAN (WGAN-GP) with four architectural contributions synthesised realistic fault windows (contextual Fréchet distance 0.141). Under a leakage-free held-out protocol, a detector trained on synthetic data alone was statistically indistinguishable from one trained on real data (paired Wilcoxon $p = 0.13$).
- **Principal contribution — a design-time instrument.** Using the generator as a controllable fault source, detector accuracy and latency were mapped across the full range of fault severity and duration — an "FDIR design data sheet" that a fixed labelled dataset cannot provide, revealing that fault duration dominates severity for detectability.

Method	Mean F1	Contextual F1
Classical baseline (z-score)	0.183	0.052
SVM — supervised	0.598	0.508
VAE — unsupervised	0.590	0.638
LSTM — unsupervised	0.743	0.836

Mean F1 across 81 real telemetry channels (point-adjust). The contextual column is the story: data-driven detectors close the gap classical FDIR cannot reach.



Detection F1 of the LSTM detector as a function of injected fault severity and duration, over 20 unseen held-out channels — a controllable characterisation of detector capability that no fixed labelled dataset can supply.

Why it matters. The work moves spacecraft FDIR toward a cognitive digital twin that learns and certifies its own fault-detection capability from operational data — directly relevant to autonomous operations, on-orbit servicing, and health management of large constellations.

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